Early Manila Hokkien: digitizing and analyzing a 17th-century Chinese-Spanish dictionary

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Starting Point

The Early Manila Hokkien (EMHo) project focuses on digitizing and analyzing a 17th-century Chinese-Spanish dictionary, the "Bocabulario de lengua sangleya por las letraz de el A.B.C." (henceforth *Bocabulario*, British Library Add ms. 25.317, see Figure 1). This collaborative project is funded by DFG and FWF as part of a WEAVE initiative and brings together experts from different domains to carry out multidisciplinary research on the manuscript (Döhla et al. 2022; Döhla et al. 2024). This includes *Linguistics* with respect to Romance and Chinese languages and history, missionary linguistics, language contact and migration, and *Digital Humanities* in relation to complex text annotation and digital scholarly edition (DSE).

The project entails understanding the status of Early Manila Hokkien, the linguistic methods used by missionaries, and the socio-historical context of early Chinese migration and Spanish colonialism in Manila (Klöter / Döhla 2022; Döhla 2025). The transmission history of the manuscript is not entirely clear; it is assumed that the pocket dictionary was used as a practical digest and was copied from a larger text by non-Spanish "copyists who do not seem to have grown up with the European tradition of scripturality" (Döhla 2025). The project emphasizes careful transcription and annotation, particularly of linguistic and tonal features both following and expanding the Text Encoding Initiative (TEI) standard. Related work for the application of TEI to multilingual, multiscript, and non-European language documentation includes among others Bowers (2020), Czaykowska-Higgins et al. (2014), Ngué Um (2017), and VedaWeb (2024). The establishment of various working and interest groups on multilingual issues highlights the importance of this topic (cf. https://dig-hum.de/ag-multilingual-dh , https://adho.org/sigs/#multilingual-DH , https://multilingual.hypotheses.org/ , https://tei-c.org/activities/sig/eastasian/).

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    <form type="lemma" xml:lang="nan-Latn-PH">Kei</form>.
    <sense xml:lang="es-Latn-PH">
        <choice>
            <abbr>Sig<am>.</am><hi rend="superscript">ca</hi></abbr>
            <expan>Sig<ex>nifi</ex>ca</expan>
        </choice>
        <def>
           confe<choice><orig>ç</orig><reg>s</reg></choice>ar
        </def><supplied>;</supplied>
        <choice>
              <orig><choice><abbr>V.g.</abbr><expan>verbi gratia</expan></choice></orig>
              <reg><choice><abbr>p.ej.</abbr><expan>por ejemplo</expan></choice></reg>
        </choice>
        <cit type="example" xml:lang="nan-Latn-PH" xml:id="entry.0738.ex.1">
            <quote>Lu<supplied> </supplied>ay Kei<surplus>.</surplus><lb/> Chue.</quote>
            <cit type="translation" xml:lang="es-Latn-PH">
               <quote>qui<choice><orig>e</orig><reg>é</reg></choice>res
                   <surplus>. </surplus>te confe<choice><orig>ç</orig>
                   <reg>s</reg></choice>ar
               </quote>
            </cit>
        </cit>
    </sense>
</entryFree>
<entry type="translation" corresp="#entry.0738">
    <form type="lemma">
        <orth xml:lang="nan-Latn-tailo">Ké</orth>
        <orth xml:lang="nan-Hani-PH">解<note n="1"><p>[...]</p></note></orth>
    </form>
    <sense xml:lang="en-Latn"> means <def>to confess</def>; e.g.
        <cit type="example" xml:lang="nan-Latn-tailo" corresp="#entry.0738.ex.1">
            <quote>lú ài ké tsuē</quote>
            <cit type="translation" xml:lang="nan-Hani-PH">
               <quote>你愛解罪</quote>
              </cit>
              <cit type="translation" xml:lang="en-Latn">
                 <quote>you want to confess yourself<note n="2"><p>[...]</p></note>
                 </quote>
              </cit>
         </cit>
    </sense>
</entry>
```

Add ms. 25.317, fol. 125v. Facsimile reused with kind permission © British Library Board, all rights reserved.

Objectives and DSE

The dictionary offers valuable insights into the Southern Min language, also known as Hokkien, as spoken by Chinese immigrants in early Ma-

nila. The primary objective is to create a DSE of this historical document on multiple levels (see Figure 2). It comprises digital facsimiles of the original folios, a diplomatic transcription, a critical edition, as well as an English translation. Additionally, the edition will include linguistic annotations, such as morpho-syntactic analysis, tone (Maddieson 2013; Yip 2002), and Chinese characters, to benefit a broader linguistic audience. The DSE will offer clear distinctions between the complex annotation levels of the manuscript and provide multiple tailor-made synoptic views of the resource. Building on the work of Bowers (2020), the project seeks to provide a recommendation to the TEI Guidelines for the annotation of tone, which would be a valuable service to the community and could help diversify the use of the guidelines.

```
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    <sense xml:lang="es-Latn-PH">
        <choice>
            <abbr>Sig<am>.</am><hi rend="superscript">ca</hi></abbr>
            <expan>Sig<ex>nifi</ex>ca</expan>
        </choice>
        <def>
           confe<choice><orig>ç</orig><reg>s</reg></choice>ar
        </def><supplied>;</supplied>
        <choice>
              <orig><choice><abbr>V.g.</abbr><expan>verbi gratia</expan></choice></orig>
              <reg><choice><abbr>p.ej.</abbr><expan>por ejemplo</expan></choice></reg>
        </choice>
        <cit type="example" xml:lang="nan-Latn-PH" xml:id="entry.0738.ex.1">
            <quote>Lu<supplied> </supplied>ay Kei<surplus>.</surplus><lb/> Chue.</quote>
            <cit type="translation" xml:lang="es-Latn-PH">
               <quote>qui<choice><orig>e</orig><reg>é</reg></choice>res
                   <surplus>. </surplus>te confe<choice><orig>ç</orig>
                   <reg>s</reg></choice>ar
               </quote>
            </cit>
        </cit>
    </sense>
</entryFree>
<entry type="translation" corresp="#entry.0738">
    <form type="lemma">
        <orth xml:lang="nan-Latn-tailo">Ké</orth>
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              </cit>
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                 <quote>you want to confess yourself<note n="2"><p>[...]</p></note>
                 </quote>
              </cit>
         </cit>
    </sense>
</entry>
```

Multiple levels of the DSE.

The larger aim of the project is to contribute to the digital documentation and open-access publication of non-European language resources, offering fresh perspectives on Chinese-Spanish lexicography and the history of the Chinese language, with a specific focus on marginalized groups.

Preliminary Results

The workflow of data capture from transcription to annotation is of central concern in this first project phase, which explores suitable (semi-)automatic assistance and quality control mechanisms to ensure smooth cooperation between partners with different research questions and domain backgrounds as well as physical locations. Strategies for tackling challenges like coordinating simultaneous work on different edition levels or languages and scripts of the manuscript and keeping track of versions and changes are thus needed as well. A preliminary modeling approach to one dictionary entry is discussed below (shortened and simplified).

```
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    <form type="lemma" xml:lang="nan-Latn-PH">Kei</form>.
    <sense xml:lang="es-Latn-PH">
        <choice>
            <abbr>Sig<am>.</am><hi rend="superscript">ca</hi></abbr>
            <expan>Sig<ex>nifi</ex>ca</expan>
        </choice>
        <def>
           confe<choice><orig>ç</orig><reg>s</reg></choice>ar
        </def><supplied>;</supplied>
        <choice>
              <orig><choice><abbr>V.g.</abbr><expan>verbi gratia</expan></choice></orig>
              <reg><choice><abbr>p.ej.</abbr><expan>por ejemplo</expan></choice></reg>
        </choice>
        <cit type="example" xml:lang="nan-Latn-PH" xml:id="entry.0738.ex.1">
            <quote>Lu<supplied> </supplied>ay Kei<surplus>.</surplus><lb/> Chue.</quote>
            <cit type="translation" xml:lang="es-Latn-PH">
               <quote>qui<choice><orig>e</orig><reg>é</reg></choice>res
                   <surplus>. </surplus>te confe<choice><orig>ç</orig>
                   <reg>s</reg></choice>ar
               </quote>
            </cit>
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              </cit>
              <cit type="translation" xml:lang="en-Latn">
                 <quote>you want to confess yourself<note n="2"><p>[...]</p></note>
                 </quote>
              </cit>
         </cit>
    </sense>
</entry>
```

Diplomatic transcription, critical edition, and English translation of one dictionary entry.

The TEI XML heavily relies on the TEI Lex-0 recommendations for encoding dictionaries (Tasovac et al. 2018, TEI Consortium 2024) combined with editorial and other alterations to the text. In Code Example 1 the <entryFree> element preserves each dictionary entry in its original form, representing the diplomatic and critical edition. The corresponding <entry> element supplements this with translations into English and modern transliteration (e.g. Tâi-lô). The constant change between languages and scripts is encoded by the @xml:lang attribute.

The implementation of the linguistic analysis level is yet to be modeled but will require structuring examples into smaller units and integrating interlinear glossing information following the Leipzig Glossing Rules (EVA MPG 2015) and the inclusion of tonal categorization, which could

be encoded into feature structure elements (<fs>) (Steiner et al. 2022).

Conclusion

By making a significant historical document accessible, the EMHo project contributes to linguistics and digital humanities by creating a multi-layered DSE that documents Early Manila Hokkien's linguistic features, examines Spanish missionary approaches to Chinese language documentation and offers considerations on text encoding for representing historical and under-resources languages.

The contribution will present preliminary results from the first project year. This includes the conceptual definition of the digital edition's various levels as well as the corresponding approaches to text and data modeling, keeping in mind the respective target group and striving for maximum data interoperability. Both aspects are key to enabling the reuse of this unique source by different domains and scholarly communities.

Offering sources like the *Bocabulario* as open research data highlights that the usage of multiple languages and scripts is and was not a marginal case but rather the norm. Leveraging the potential of such sources can empower respective communities (cf. Carroll et al. 2020) and offer new perspectives on past and present language contact scenarios.

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